

ST5209/X Assignment 3

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Submission

1. Render the document to get a .pdf printout.
2. Submit both the .qmd and .pdf files to Canvas.

1. Holt-Winters, residuals, and forecast accuracy

Consider the antidiabetic drug sales time series which can be loaded using the following code snippet.

```
diabetes <- read_rds("../_data/cleaned/diabetes.rds") |>
  select(TotalC)
```

- a. Fit the following exponential smoothing models on the entire time series:

```
fit <- diabetes |>
  model(
    hw_mult = ETS(TotalC ~ error("M") + trend("A") + season("M")),
    hw_log = ETS(log(TotalC) ~ error("A") + trend("A") + season("A")),
    hw_mult_damped = ETS(TotalC ~ error("M") + trend("Ad") + season("M"))
  )
fit
```

```
# A mable: 1 x 3
      hw_mult      hw_log hw_mult_damped
  <model>      <model>      <model>
1 <ETS(M,A,M)> <ETS(A,A,A)> <ETS(M,Ad,M)>
```

- b. Make ACF plots for the innovation residuals of these three models.

```

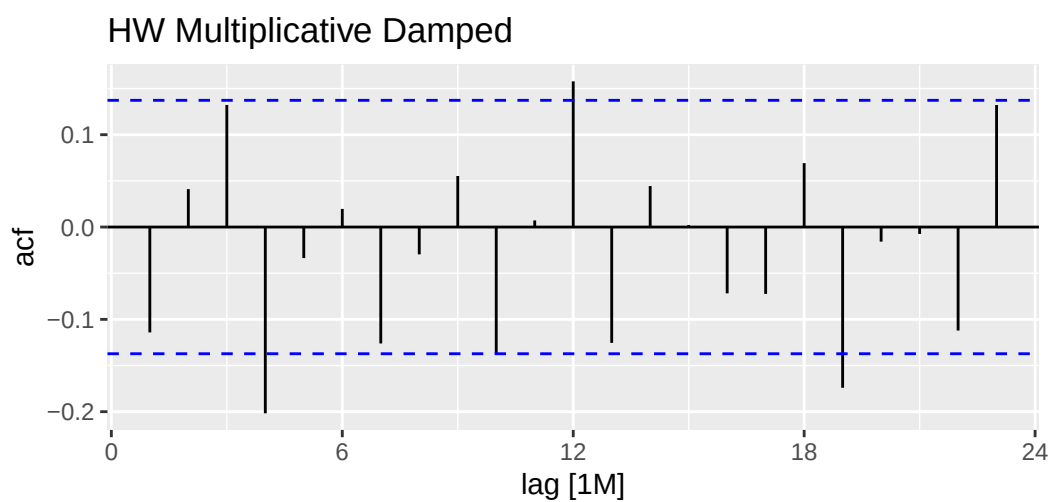
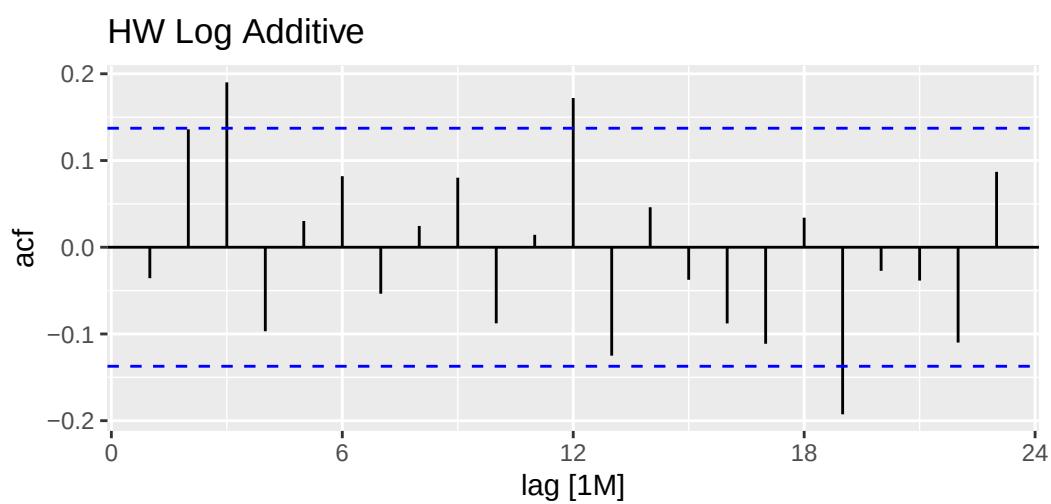
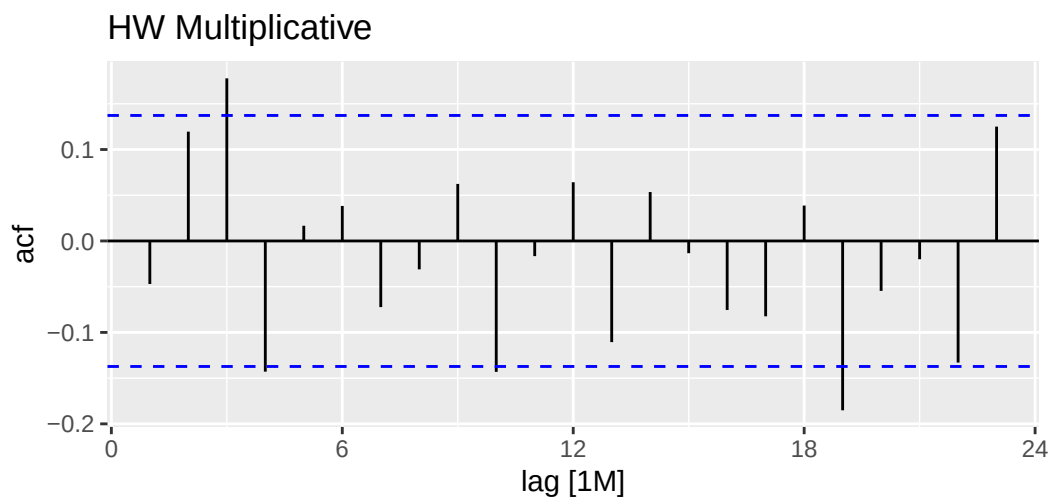
library(gridExtra)

resid_data <- fit |> augment()

p1 <- resid_data |> filter(.model == "hw_mult") |>
  ACF(.innov) |> autoplot() + ggtitle("HW Multiplicative")
p2 <- resid_data |> filter(.model == "hw_log") |>
  ACF(.innov) |> autoplot() + ggtitle("HW Log Additive")
p3 <- resid_data |> filter(.model == "hw_mult_damped") |>
  ACF(.innov) |> autoplot() + ggtitle("HW Multiplicative Damped")

grid.arrange(p1, p2, p3, nrow = 3)

```



From the ACF plots, we can check whether the innovation residuals resemble white noise by seeing if most autocorrelations fall within the blue significance bands. Any significant spikes outside the bands suggest remaining structure not captured by the model.

- c. Calculate the p-value from a Ljung-Box test on the residuals with lag $h = 8$.

```
fit |>
  augment() |>
  features(.innov, ljung_box, lag = 8)
```

```
# A tibble: 3 x 3
  .model      lb_stat lb_pvalue
  <chr>      <dbl>    <dbl>
1 hw_log      16.0     0.0424
2 hw_mult     16.0     0.0420
3 hw_mult_damped 19.2     0.0141
```

The Ljung-Box test null hypothesis is that the residuals are white noise. A p-value > 0.05 means we fail to reject and the model fit is adequate; a p-value < 0.05 indicates significant residual autocorrelation, meaning the model has not fully captured the time series dynamics and could be improved.

- d. Perform time series cross-validation for the three methods.

```
diabetes_cv <- diabetes |>
  stretch_tsibble(.init = 50, .step = 10)

cv_fit <- diabetes_cv |>
  model(
    hw_mult = ETS(TotalC ~ error("M") + trend("A") + season("M")),
    hw_log = ETS(log(TotalC) ~ error("A") + trend("A") + season("A")),
    hw_mult_damped = ETS(TotalC ~ error("M") + trend("Ad") + season("M"))
  )

cv_fc <- cv_fit |>
  forecast(h = 4) |>
  group_by(.id, .model) |>
  mutate(h = row_number()) |>
  ungroup() |>
  as_fable(response = "TotalC", distribution = TotalC)

cv_fc |>
```

```
accuracy(diabetes) |>
select(.model, RMSSE, MASE) |>
arrange(RMSSE)
```

```
# A tibble: 3 x 3
  .model      RMSSE  MASE
  <chr>      <dbl> <dbl>
1 hw_log      0.655 0.592
2 hw_mult_damped 0.659 0.622
3 hw_mult      0.683 0.630
```

The model with the smallest RMSSE provides the best forecast accuracy among the three. With `.init = 50` and `.step = 10`, the number of CV folds is $\lfloor (n - 50)/10 \rfloor + 1$. For each fold, 4 forecast errors are computed (one per horizon), so the total number of data points the error is averaged over is (number of folds) $\times$ 4.

2. ETS parameters, prediction intervals

The dataset `hh_budget` contains annual indicators of household budgets for a few countries.

- a. Fit ETS on the time series comprising savings as a portion of household budgets in Canada.

```
canada_savings <- hh_budget |>
  filter(Country == "Canada")

fit_ets <- canada_savings |>
  model(ets = ETS(Savings))
fit_ets
```

```
# A mable: 1 x 2
# Key:      Country [1]
  Country      ets
  <chr>        <model>
1 Canada <ETS(A,N,N)>
```

- b. Which ETS model was selected?

```
report(fit_ets)
```

```

Series: Savings
Model: ETS(A,N,N)
Smoothing parameters:
  alpha = 0.9998999

Initial states:
  l[0]
9.310387

sigma^2: 1.3942

      AIC      AICc      BIC
79.21722 80.55055 82.49035

```

The automatically selected ETS model is shown above. The selection is based on minimizing AICc.

c. What are the fitted parameters of the model?

```
fit_ets |> tidy()
```

```

# A tibble: 2 x 4
  Country .model term   estimate
<chr>    <chr> <chr>    <dbl>
1 Canada ets   alpha    1.000
2 Canada ets   l[0]     9.31

```

d. Based on the fitted parameters, what simple forecasting method is this similar to?

```

params <- fit_ets |> tidy()
alpha_val <- params |> filter(term == "alpha") |> pull(estimate)
cat("alpha =", round(alpha_val, 4), "\n")

```

```
alpha = 0.9999
```

```

if (alpha_val > 0.8) {
  cat("Alpha is close to 1, so the model is similar to the naive method.\n")
} else if (alpha_val < 0.2) {
  cat("Alpha is close to 0, so the model is similar to the mean method.\n")
} else {
  cat("Alpha is moderate, so the model is a genuine blend between naive and mean methods.\n")
}

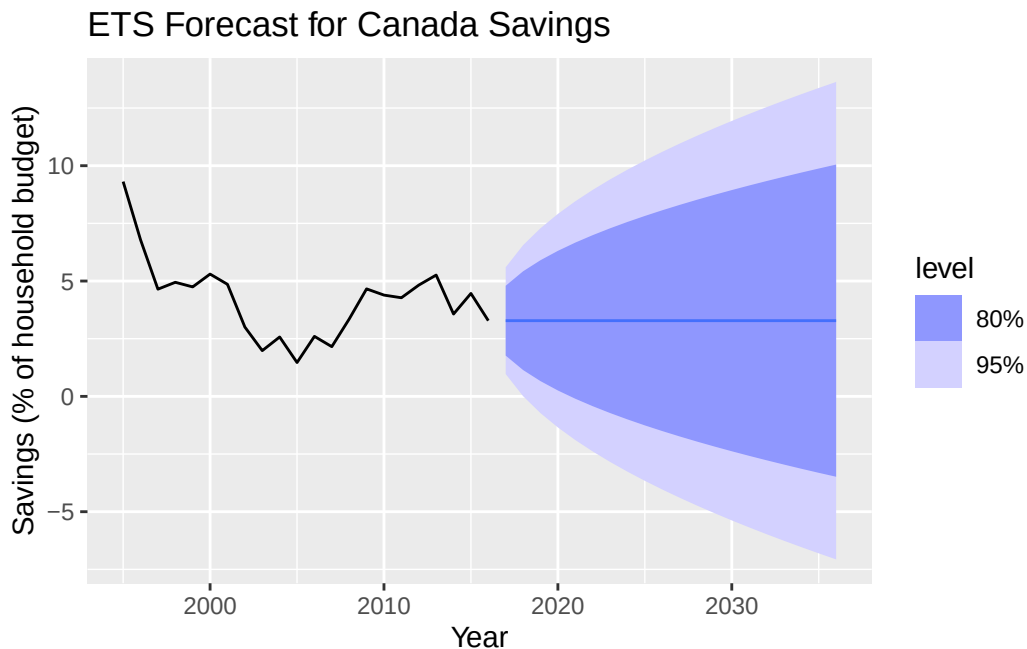
```

Alpha is close to 1, so the model is similar to the naive method.

For SES, $\alpha \rightarrow 1$ gives the naive method (forecast = last observation), while $\alpha \rightarrow 0$ gives the mean method (forecast = average of all observations).

- e. Based on your answer to d), how does the width of the prediction interval change as a function of the forecast horizon h ?

```
fit_ets |>
  forecast(h = 20) |>
  autoplot(canada_savings) +
  labs(title = "ETS Forecast for Canada Savings",
       y = "Savings (% of household budget)")
```



For SES (ETS(A,N,N)), the prediction interval width is proportional to $\sqrt{1 + (h - 1)\alpha^2}$. When $\alpha \approx 1$ (naive), the width grows like $\sqrt{h}$; when $\alpha \approx 0$ (mean), the width is approximately constant. The forecast plot above verifies this behavior.

3. Moving averages and differences

Consider the linear trend model

$$X_t = \beta_0 + \beta_1 t + W_t.$$

Define a time series (Y_t) by taking a moving average of (X_t) with a symmetric window of size 7. Define another time series (Z_t) by taking a difference of (X_t) .

a. **Mean function for (Y_t) :**

$$Y_t = \frac{1}{7} \sum_{j=-3}^3 X_{t+j} = \frac{1}{7} \sum_{j=-3}^3 (\beta_0 + \beta_1(t+j) + W_{t+j})$$

$$\mu_Y(t) = E[Y_t] = \frac{1}{7} \sum_{j=-3}^3 (\beta_0 + \beta_1(t+j)) = \beta_0 + \beta_1 t + \frac{\beta_1}{7} \sum_{j=-3}^3 j = \beta_0 + \beta_1 t$$

since $\sum_{j=-3}^3 j = 0$. The mean function depends on t , so (Y_t) is **not stationary**.

ACVF for (Y_t) :

$$Y_t - \mu_Y(t) = \frac{1}{7} \sum_{j=-3}^3 W_{t+j}$$

$$\gamma_Y(s, t) = \text{Cov}(Y_s, Y_t) = \frac{1}{49} \sum_{i=-3}^3 \sum_{j=-3}^3 \text{Cov}(W_{s+i}, W_{t+j}) = \frac{\sigma^2}{49} \cdot |\{(i, j) : s+i = t+j, -3 \leq i, j \leq 3\}|$$

Let $h = t - s$. The number of overlapping terms is $\max(7 - |h|, 0)$. Thus:

$$\gamma_Y(h) = \frac{\sigma^2}{49} \max(7 - |h|, 0), \quad |h| \leq 6$$

The ACVF depends only on $h = t - s$, not on s or t individually.

b. **Mean function for (Z_t) :**

$$Z_t = X_t - X_{t-1} = \beta_1 + W_t - W_{t-1}$$

$$\mu_Z = E[Z_t] = \beta_1$$

The mean is constant (does not depend on t).

ACVF for (Z_t) :

$$Z_t - \mu_Z = W_t - W_{t-1}$$

$$\gamma_Z(h) = \text{Cov}(W_t - W_{t-1}, W_{t+h} - W_{t+h-1}) = \begin{cases} 2\sigma^2 & h = 0 \\ -\sigma^2 & |h| = 1 \\ 0 & |h| \geq 2 \end{cases}$$

(Z_t) is **stationary** since both the mean and ACVF do not depend on t .

c. **CCF of (Y_t) and (Z_t) :**

$$\begin{aligned} \gamma_{YZ}(s, t) &= \text{Cov}(Y_s, Z_t) = \text{Cov}\left(\frac{1}{7} \sum_{j=-3}^3 W_{s+j}, W_t - W_{t-1}\right) \\ &= \frac{1}{7} \sum_{j=-3}^3 [\text{Cov}(W_{s+j}, W_t) - \text{Cov}(W_{s+j}, W_{t-1})] \\ &= \frac{\sigma^2}{7} [\mathbf{1}(|t-s| \leq 3) - \mathbf{1}(|t-1-s| \leq 3)] \end{aligned}$$

Writing $h = t - s$, this becomes:

$$\gamma_{YZ}(h) = \frac{\sigma^2}{7} [\mathbf{1}(|h| \leq 3) - \mathbf{1}(|h-1| \leq 3)]$$

The cross-covariance itself only depends on the lag h . However, since (Y_t) is not stationary, the CCF (which requires joint stationarity) is not well-defined in the usual sense.

d. **Are (Y_t) and (Z_t) jointly stationary?**

No. (Y_t) has a mean function $\mu_Y(t) = \beta_0 + \beta_1 t$ that depends on t , so (Y_t) is not (marginally) stationary. Since joint stationarity requires both marginal processes to be stationary, (Y_t, Z_t) cannot be jointly stationary.

4. Sample vs population ACF

Consider the signal plus noise model

$$X_t = \sin(2\pi t/5) + W_t.$$

a. **ACF of (X_t) :**

The mean function is $\mu_X(t) = E[X_t] = \sin(2\pi t/5)$, which depends on t . Therefore (X_t) is **not weakly stationary**, and the ACF $\rho(h)$ as a function of lag alone is **not well-defined**.

We can still compute the autocovariance function:

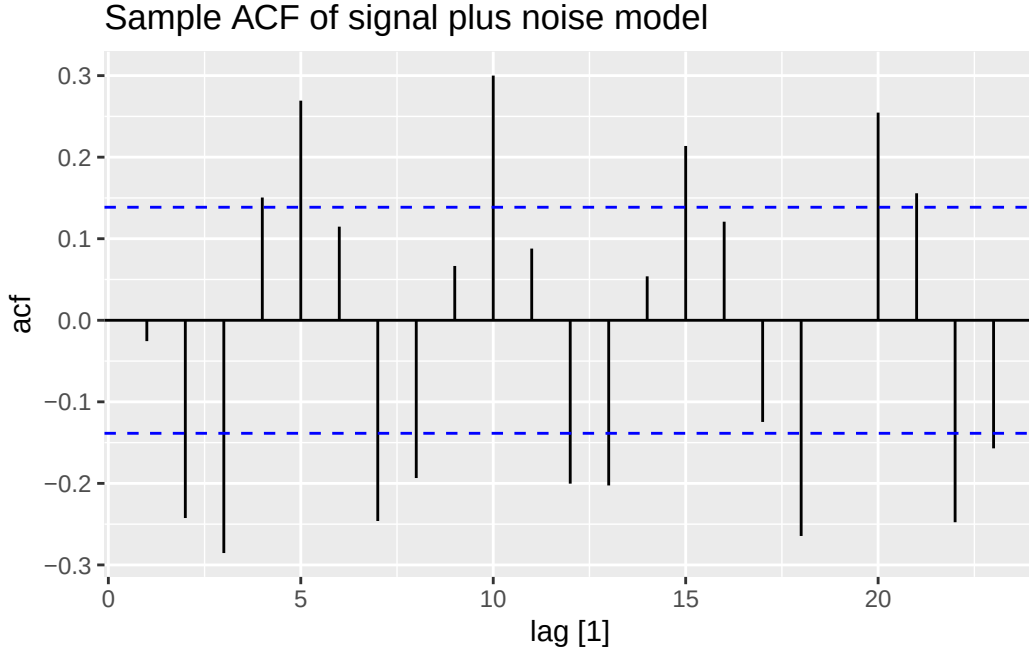
$$\gamma_X(s, t) = \text{Cov}(X_s, X_t) = \text{Cov}(W_s, W_t) = \begin{cases} \sigma^2 & s = t \\ 0 & s \neq t \end{cases}$$

Although this depends only on $s - t$, the non-constant mean means the process fails the stationarity requirement. The standard ACF framework does not apply.

b. Simulate and plot the sample ACF.

```
set.seed(5209)
n <- 200
signal_noise <- tibble(
  t = 1:n,
  X = sin(2 * pi * (1:n) / 5) + rnorm(n)
) |>
  as_tsibble(index = t)

signal_noise |>
  ACF(X) |>
  autoplot() +
  labs(title = "Sample ACF of signal plus noise model")
```



c. Why does the sample ACF not look like the population ACF?

The sample ACF uses the sample mean $\bar{X}$ to center the data. However, the population mean $\mu_X(t) = \sin(2\pi t/5)$ varies with t , so $\bar{X}$ is not a good estimate of the individual means. The centering by $\bar{X}$ does not remove the sinusoidal signal, causing the sample ACF to show periodic patterns that reflect the signal rather than the true (zero) autocorrelation of the noise.

d. Why does the asymptotic normality theorem for the ACF not apply?

The asymptotic normality theorem requires the time series to be stationary. Since (X_t) has a mean function $\mu_X(t) = \sin(2\pi t/5)$ that depends on t , the process is not stationary, and the theorem does not apply.

5. Gaussian processes

Consider a random vector (X_1, X_2, X_3, X_4) with the joint Gaussian distribution

$$\begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix} \sim N \left(\begin{bmatrix} -0.5 \\ 0 \\ 0.5 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 & 1 & 1 & 1 \\ 1 & 3 & 1 & 1 \\ 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 3 \end{bmatrix} \right)$$

```

Sigma <- matrix(c(3, 1, 1, 1,
                  1, 3, 1, 1,
                  1, 1, 3, 2,
                  1, 1, 2, 3),
                nrow = 4, ncol = 4, byrow = TRUE)

mu <- c(-0.5, 0, 0.5, 2)

```

a. **Marginal variance of X_4 :**

```
Sigma[4, 4]
```

```
[1] 3
```

The marginal variance of X_4 is $\Sigma_{44} = 3$.

b. **Conditional variance of X_4 given $X_1 = 1, X_2 = 1, X_3 = 1$:**

Using the Gaussian conditional formula: $\text{Var}(X_4|X_{1:3}) = \Sigma_{44} - \Sigma_{4,1:3}\Sigma_{1:3,1:3}^{-1}\Sigma_{1:3,4}$

```

Sigma_11 <- Sigma[1:3, 1:3]
Sigma_12 <- Sigma[1:3, 4]
Sigma_21 <- Sigma[4, 1:3]
Sigma_22 <- Sigma[4, 4]

cond_var <- Sigma_22 - Sigma_21 %*% solve(Sigma_11) %*% Sigma_12
cond_var

```

```

      [,1]
[1,]  1.6

```

The conditional variance is $3 - [1, 1, 2]\Sigma_{1:3}^{-1}[1, 1, 2]^T = 3 - 7/5 = 8/5 = 1.6$. It does **not** depend on the observed values $X_1 = 1, X_2 = 1, X_3 = 1$, since in a Gaussian distribution, the conditional variance depends only on the covariance structure, not on the conditioning values.

c. **Conditional mean of X_4 given $X_1 = 1, X_2 = 1, X_3 = 1$:**

Using the formula: $E(X_4|X_{1:3} = x) = \mu_4 + \Sigma_{4,1:3}\Sigma_{1:3,1:3}^{-1}(x - \mu_{1:3})$

```
x_obs <- c(1, 1, 1)
mu_1 <- mu[1:3]
mu_2 <- mu[4]

cond_mean <- mu_2 + Sigma_21 %*% solve(Sigma_11) %*% (x_obs - mu_1)
cond_mean
```

```
      [,1]
[1,] 2.55
```

With $x - \mu_{1:3} = [1, 1, 1] - [-0.5, 0, 0.5] = [1.5, 1, 0.5]$, the conditional mean is $2 + [1, 1, 2]\Sigma_{1:3}^{-1}[1.5, 1, 0.5]^T = 2 + 0.55 = 2.55$.

d. **Level 95% prediction interval for X_4 :**

```
cond_sd <- sqrt(cond_var)
lower <- cond_mean - 1.96 * cond_sd
upper <- cond_mean + 1.96 * cond_sd
cat("95% Prediction Interval: [", round(lower, 4), ",", round(upper, 4), "]\n")
```

```
95% Prediction Interval: [ 0.0708 , 5.0292 ]
```

The 95% prediction interval is $2.55 \pm 1.96\sqrt{1.6} = 2.55 \pm 2.479$, i.e. $[0.071, 5.029]$.